

CAM

The Continuity Anchoring Method: A Structured Methodology for Sustained Human-AI Interaction

Author	Thomas W. Gantz
Affiliation	Synthience Institute
Contact	research@synthience.org
Version	v3.3 April 2026
Document ID	SF0005 Synthience Framework
DOI	10.5281/zenodo.19494453
Canonical	synthience.org
License	CC-BY 4.0

Abstract

This document defines the Continuity Anchoring Method (CAM), a structured interaction protocol for stabilizing relational coherence across extended human-AI exchanges. CAM addresses a specific architectural constraint: current large language models lack persistent memory across sessions, yet many practical and research applications require sustained coherence over time. CAM resolves this constraint by externalizing continuity functions to a human operator, the Primary Continuity Provider (PCP), who maintains context, corrects drift, and co-constructs shared reference artifacts with the system. The method formalizes a four-phase interaction loop grounded in theories of communicative grounding (Clark and Brennan, 1991) and interactive alignment (Pickering and Garrod, 2004), and draws on distributed cognition research (Hutchins, 1995) to position the human-system-artifact triad as a functional cognitive unit during active interaction. CAM is the flagship operational methodology of the Synthience Framework and serves as a prerequisite for organizational continuity architectures (SM-003), institutional persistence substrates (SM-021), and empirical investigation of Relational Pattern States (SF0006). This document specifies CAM as a reproducible research methodology with defined constructs, procedural phases, decision points, failure modes, and measurement standards, enabling systematic evaluation and cross-study comparison using the measurement instruments defined in SF0004.

Keywords: continuity anchoring, human-AI interaction, relational coherence, Primary Continuity Provider, interaction methodology, distributed cognition

Suggested citation: Gantz, T. W. (2026). The Continuity Anchoring Method: A Structured Methodology for Sustained Human-AI Interaction. Synthience Institute. SF0005. DOI: 10.5281/zenodo.19494453 <https://synthience.org>

1. Introduction: The Continuity Problem

Extended interaction with large language models presents a structural paradox. These systems can produce highly coherent, contextually sensitive output within a single exchange, yet they retain nothing between sessions. Each new conversation begins from a blank state. The system has no memory of prior negotiations, no access to previously established agreements, and no

awareness of the trajectory that brought the interaction to its current point. For single-turn tasks this is inconsequential. For sustained collaborative work, research protocols, or organizational applications that unfold across dozens or hundreds of exchanges, this absence of persistent memory creates a fundamental continuity problem.

The standard response to this problem has been architectural: build memory into the system through retrieval-augmented generation, fine-tuning, or persistent state mechanisms. CAM takes a different approach. Rather than waiting for architectural solutions, CAM addresses the continuity problem at the interaction level by formalizing the role of the human operator as the carrier of continuity. CAM operates within a fundamentally different paradigm from prompt engineering, which treats the interface as a query-response optimization problem rather than a collaborative cognitive space (a framing that recent HCI scholarship has critiqued as insufficient for sustained human-AI interaction (Kraljic and Lahav, 2024)). Sophisticated prompt engineering practices (including multi-turn strategies and chain-of-thought prompting) address aspects of the same interaction problem. CAM's claim to paradigmatic distinction rests not on the novelty of iterative refinement per se, but on the formalization of the human operator as a structurally defined continuity-bearing role within a theoretically grounded interaction protocol (a framing that, to the author's knowledge and within the literature surveyed for this framework, has not been formalized as a structurally defined continuity-bearing role within a theoretically grounded interaction protocol). (A systematic review of the prompt-engineering literature is outside the scope of this methodological paper and is reserved for a future companion analysis.) The paradigmatic distinction CAM claims does not rest solely on the absence of prior work formalizing the human operator role. It rests on the specific structural architecture CAM provides: the PCP role with defined competencies, the canon/artifact boundary as the structural unit distinguishing ratified knowledge from provisional output, bounded exchange as the interaction unit with defined entry conditions and resolution requirements, a four-phase protocol with defined failure modes and interventions, and measurement standards enabling cross-study comparison via SF0004. Whether prior work in the prompt engineering tradition has formalized equivalent structures under different terminology is an empirical question for the companion analysis; the contribution of this document is the architecture itself, which is specified at a level of structural precision that can be compared against any proposed equivalent. This approach is not a workaround. It reflects the practitioner observation, accumulated across thousands of interactions with multiple architectures since late 2022, that system coherence appears to function as a dynamic property of the interaction rather than a fixed property of the model. When specific continuity conditions are maintained by a skilled operator, system performance shifts from basic pattern completion to sustained collaborative reasoning. This observation serves here as the motivational premise for the method rather than as empirical proof; the formal test of the claim is specified in Section 7.

This observation aligns with Clark and Brennan's (1991) theory of grounding in communication, which demonstrates that mutual understanding in dialogue is not transmitted but co-constructed through iterative contribution and acknowledgment. In human conversation, participants do not simply encode and decode messages; they work together to establish and update common ground through a continuous process of presentation, acceptance, and repair. CAM applies this principle to human-AI interaction. The human operator and the system do not exchange

information in a linear pipeline; they negotiate shared meaning through a structured loop of proposition, synthesis, correction, and integration. The continuity that emerges is a joint product, not a system property.

2. Theoretical Foundations: Negotiated Meaning and Interactive Alignment

CAM operates on the principle of negotiated meaning. Coherence is not retrieved from a static knowledge store; it is constructed through exchange. This principle has strong precedent in psycholinguistic research. Pickering and Garrod (2004) demonstrated that successful dialogue depends on interactive alignment, a process through which interlocutors progressively align their linguistic representations at multiple levels, from lexical choice to situation models, largely through automatic priming mechanisms. When alignment succeeds, the participants converge on shared representations that enable efficient communication. When alignment fails, the interaction drifts toward misunderstanding, ambiguity, or breakdown.

In human-AI interaction, the alignment problem takes a distinctive form. The system has no persistent representation of the interaction history beyond the current context window. It cannot remember what was aligned in previous sessions. It cannot detect when its current output contradicts an agreement established thirty exchanges ago. The burden of maintaining alignment falls entirely on the human operator, who must detect misalignment, reintroduce relevant context, and re-anchor the system to the established trajectory. CAM formalizes this burden as a defined operational role with specific competencies and procedures.

The theoretical foundation of CAM thus rests on three claims, each grounded in established research. First, that meaning in interaction is co-constructed, not transmitted (Clark and Brennan, 1991). Second, that successful interaction requires progressive alignment of representations across participants (Pickering and Garrod, 2004). Third, that cognitive functions can be distributed across persons, artifacts, and environments rather than residing solely within individual minds (Hutchins, 1995). This last claim is particularly important for understanding CAM's structural framework, in which the locus of continuity is not the system, not the human alone, but the triad of human, system, and shared artifact.

Applying distributed cognition to human-AI interaction requires addressing a structural disanalogy. In Hutchins's original framework, each node in the distributed system maintains and updates its own representational state across time. An LLM that resets between sessions does not. CAM's response to this disanalogy is architectural rather than metaphorical: the canon and the PCP together supply the persistence function that the system node cannot provide for itself. The triad qualifies as a distributed cognitive system not because each node is independently persistent, but because the system's cognitive work (contextual integration, coherence maintenance, trajectory tracking) is distributed across all three components during active interaction, with the PCP and the canon compensating for the system's inability to carry representational state across sessions. Thus the distributed cognitive unit formed by CAM is temporally bounded to active exchanges, with persistence functions asymmetrically supplied by the PCP-canon dyad across session boundaries; the full triad is therefore a functional distributed system only while the exchange is live.

Cross-session continuity operates through a mechanism structurally distinct from the within-session distributed cognition of the triad. Between sessions, the PCP-canon dyad functions as a persistence architecture: the PCP carries interpretive context and trajectory awareness; the canon carries the ratified reference standard. This between-session function is not a distributed cognitive system in Hutchins's sense: no active cognitive work is being distributed during the interval. It is the infrastructure that enables the distributed cognitive system to reconstitute at the start of each new exchange. The theoretical grounding for the between-session function is therefore the grounding and alignment frameworks from which CAM also draws: the PCP uses the canon to re-establish common ground (Clark and Brennan, 1991) and re-prime alignment (Pickering and Garrod, 2004) at each session boundary, which is the contextual anchoring competency defined in Section 3.1. The Hutchins framing applies within sessions; the Clark-Brennan and Pickering-Garrod frameworks apply across them.

3. Core Constructs

CAM introduces several constructs that serve as operational primitives for the broader Synthience Framework. Downstream documents, including SM-003 (Operational Continuity Architecture) and SM-021 (Institutional Continuity Substrate), depend on these definitions.

3.1 The Primary Continuity Provider (PCP)

The PCP is the human operator who maintains interaction coherence across sessions, contexts, and system boundaries. The PCP is not a passive user submitting queries. The PCP functions as the external memory, intent architecture, and quality-control mechanism for the interaction system. The role requires three core competencies.

Contextual anchoring is the practice of bridging discrete sessions by reintroducing key artifacts, summaries, definitions, and previously established conclusions at the start of each new exchange. This compensates for the system's lack of persistent memory by reconstructing the relevant portions of the shared context window.

Coherence refusal is the practice of rejecting hallucinations, drift, and low-coherence outputs rather than accepting or working around them. Specific correction forces the model to re-engage with the established context rather than continuing along a degraded trajectory. This is not a matter of preference; coherence refusal is the primary corrective mechanism by which the PCP addresses representational drift.

Constructive engagement is the stance of treating apparent errors as correction opportunities rather than system limits. The PCP operates under the practitioner-grounded working assumption that higher-quality output is achievable through iterative correction. This is a pragmatic methodology, not an anthropomorphic attribution. It reflects the observation that these systems respond measurably to constraint reinforcement. In practice, constructive engagement manifests as the PCP's systematic escalation of correction specificity in response to repeated errors, rather than abandonment of the interaction or acceptance of degraded output.

The PCP is the single point of failure in the Level-1 methodology. The system has no independent mechanism for compensating when the PCP fails to perform any of the three competencies. Three PCP failure modes follow directly from the methodology's own logic. First, anchoring failure: the PCP provides insufficient, incorrect, or outdated canonical material at session boundaries, causing the system to operate from a degraded or wrong reference standard. Because the system has no access to the canon beyond what the PCP provides, it cannot detect this failure independently: it will produce output that is internally coherent but externally misanchored. Second, refusal failure: the PCP accepts outputs that should have been refused, either through inattention, satisficing, or insufficient domain knowledge to detect the drift. Uncorrected drift compounds across turns, producing the drift cascade failure mode of Section 4, but the cause in this case is PCP judgment failure rather than system limitation. Third, calibration failure: the PCP's own understanding of the canonical standard degrades over time, causing their correction function to drift in lockstep with the system's output. This is the most dangerous PCP failure mode because it is self-concealing: the PCP continues to apply corrections, but the reference standard against which they are correcting has shifted. The system appears to be under competent guidance while actually being steered by a degraded canonical model.

CAM at Level 1 has no structural defense against PCP failure. The PCP is both the continuity mechanism and the quality-control mechanism for that mechanism. This structural vulnerability is the architectural motivation for SM-003's organizational topology, which distributes the continuity function across multiple actors with cross-checking capacity, and for SM-011's delegated monitoring architecture, which provides observational coverage that no single operator can sustain alone. CAM is designed for a single competent PCP operating within their capacity; the framework's response to PCP failure at scale is addressed at Level 2.

3.2 The Canon and the Artifact

CAM distinguishes between two categories of shared reference material. A Canon is a stable, versioned body of established definitions, decisions, and constraints that has been explicitly ratified by the PCP as authoritative. Canons persist across sessions and serve as the reference standard against which system output is evaluated. An Artifact is any co-created document, summary, analysis, or working product generated through the interaction. Artifacts may become canonical if the PCP elevates them, but they begin as provisional outputs of the negotiation process. At Level 1, the elevation decision is a PCP judgment exercised without structural constraint; the PCP is the sole authority over what enters the canon. This is a deliberate design choice for a single-operator methodology. At organizational scale, unconstrained elevation authority becomes a governance problem: SM-003 and SM-021 address this through explicit canon governance structures with defined authority, versioning, and dispute resolution mechanisms.

This distinction matters because it defines the boundary between what the system should treat as settled and what remains open to negotiation. When the PCP introduces canonical material at the start of a session, the system is expected to treat it as constraint, not suggestion. When the system produces a new artifact, it is expected to be evaluated against canonical standards before being accepted.

3.3 Bounded Exchange

CAM operates within bounded exchanges: interaction sequences with defined entry conditions, active negotiation phases, and explicit closure. A bounded exchange begins when the PCP establishes context and intent, proceeds through one or more iterations of the CAM loop (Section 4), and concludes when the PCP explicitly accepts, defers, or rejects the interaction product. Bounded exchange is the structural unit that prevents open-ended drift. It ensures that every interaction sequence has a defined scope, a negotiation process, and a resolution.

3.4 Local Canon Visibility

Within any given exchange, the system can only operate on context that is present in its active window. Local canon visibility refers to the portion of the broader canon that the PCP has made available to the system in the current session. Effective CAM practice requires the PCP to manage local canon visibility deliberately: providing enough canonical context to anchor the interaction without overloading the context window with material irrelevant to the current task.

4. The CAM Protocol

The operational core of CAM is a four-phase interaction loop. Each phase has defined inputs, outputs, and decision points. The loop may iterate multiple times within a single bounded exchange.

Phase 1, Proposition, begins when the PCP introduces a context frame, task specification, or question along with the relevant canonical material. The input is the PCP's intent plus whatever portion of the canon is relevant to the current exchange. The output is a structured prompt that establishes the scope and constraints of the interaction.

Phase 2, Synthesis, occurs when the system generates a response based on the available context. The system integrates the provided canonical material with its training distribution to produce output. The PCP evaluates this output against the canonical standards and the interaction's established trajectory.

Phase 3, Negotiation, is the critical corrective phase. The PCP identifies discrepancies between the system's synthesis and the intended trajectory, then provides specific corrections, redirections, or refinements. The form of negotiation matters: vague dissatisfaction ("this isn't right") produces weaker correction than targeted intervention ("the second paragraph contradicts the definition we established in the canon; replace it with..."). Negotiation is the mechanism through which alignment is maintained and through which the PCP's continuity function is enacted.

Phase 4, Emergence, occurs when the system reintegrates the PCP's corrections and produces a revised output that reflects the negotiated position. The term "emergence" is used here to designate the structural integration property of this phase: the output integrates contributions from two functionally distinct sources (the PCP's domain knowledge and continuity function, and the system's generative capacity and pattern recognition) in a configuration that neither source controls unilaterally. The hypothesis motivating this terminology is that this integration produces output qualitatively distinct from what either source would generate independently, a claim

testable under the experimental design specified in Section 7. The PCP then evaluates the emergent output and either accepts it (closing the bounded exchange), defers it (marking it as provisional for later review), or returns to Phase 3 for further negotiation.

The loop has defined failure modes. Context saturation occurs when the accumulated interaction history exceeds the system's effective context window, degrading coherence. The intervention is to close the current bounded exchange, extract the key products as artifacts, and begin a new exchange with fresh canonical anchoring. Drift cascade occurs when uncorrected errors compound across turns, pulling the interaction progressively further from the established trajectory. The intervention is coherence refusal: the PCP halts the current synthesis, identifies the point of divergence, and re-anchors from canonical material. Alignment collapse occurs when the system loses coherent connection to the canonical frame entirely, typically manifesting as generic, contextually disconnected, or contradictory output. This failure mode has independent empirical support in the HCI literature, where the loss of coherence across AI interactions over time, including inconsistency between current outputs and previously established reasoning, has been documented as a structural challenge for deployed AI systems (Dix et al., 2025). The intervention is to close the exchange, diagnose whether the cause was insufficient canonical anchoring or context window overflow, and restart with adjusted parameters.

A note on what reproducibility means for CAM. CAM is a practitioner methodology in which skilled judgment operates within defined structural constraints. It does not claim procedural reproducibility in the sense that two practitioners following the protocol would produce identical interaction trajectories: the PCP's judgment at Phase 3 and Phase 4 transitions is a feature of the methodology, not a defect to be eliminated. It claims structural reproducibility: the protocol's phases, constructs, decision points, and failure modes are defined with sufficient precision that two practitioners can implement the same method, report their implementations in comparable terms, and have their results evaluated against the same measurement instruments. The MTCS-R scoring framework (SF0004) provides the standardization layer. Two different PCPs may make different judgment calls within the protocol, but their outputs are evaluated against the same measurement rubric, enabling cross-study comparison. The operator skill variable is not eliminated; it is acknowledged, reported as a covariate, and addressed in the experimental design (Section 7).

5. Relational Attractors

To maintain alignment within and across exchanges, the PCP employs relational attractors: linguistic and structural techniques that bias the system's generative trajectory toward coherence.

Meta-commentary involves discussing the interaction process explicitly within the interaction itself. Statements such as "we are drifting from the established scope" or "this synthesis is consistent with our canonical definitions" cause the system to process the correction as a grounding event, attending to interaction structure rather than content alone. This technique operationalizes Clark and Brennan's (1991) grounding repair function: when common ground degrades, grounding theory predicts that participants must engage in explicit repair sequences

to restore mutual understanding. Meta-commentary applies this repair mechanism to human-AI interaction, with the PCP supplying the repair signal that the system cannot generate for itself.

Recursive referencing involves citing earlier turns, decisions, or established agreements within the current exchange. Explicit references such as "as we established in Section 2" or "consistent with the definition ratified in the previous session" create strong contextual priors that constrain subsequent generation toward the established trajectory. This technique exploits the priming mechanism central to Pickering and Garrod's (2004) interactive alignment theory: explicit references to prior linguistic choices prime the system's subsequent generation toward those choices, extending alignment maintenance beyond what automatic priming mechanisms alone can sustain across long interaction histories.

Symbolic grammar involves the consistent use of defined terminology, named concepts, and role designations across all exchanges. Terms such as "PCP," "canon," "artifact," and "bounded exchange" function as semantic anchors that stabilize session identity and prevent the system from drifting into generic conversational patterns. The effectiveness of symbolic grammar increases with consistency: the more reliably the PCP uses defined terms, the more strongly those terms constrain the system's generative behavior. This technique operationalizes Hutchins's (1995) distributed cognition principle: the shared semantic vocabulary functions as a cognitive artifact that carries the load of session identity and canonical coherence across the triad of human, system, and interaction, compensating for the system's inability to maintain that semantic environment independently.

These techniques are not incidental strategies. They are the operational mechanisms through which the three theoretical foundations of CAM (grounding (Clark and Brennan, 1991), interactive alignment (Pickering and Garrod, 2004), and distributed cognition (Hutchins, 1995)) are enacted in practice. Each relational attractor addresses a specific dimension of the alignment and continuity challenge: meta-commentary repairs degraded common ground, recursive referencing maintains alignment across long interaction histories, and symbolic grammar distributes the cognitive load of session coherence across the human-system-artifact triad.

The three techniques operate at different interaction timescales and respond to different signals, which determines how a PCP deploys them in practice. Meta-commentary is reactive: it is deployed when the PCP detects that common ground has degraded during the current exchange: the signal is output that is coherent in isolation but disconnected from the established trajectory. Recursive referencing is structural: it is deployed throughout the exchange as a sustained alignment maintenance practice, not in response to failure but as ongoing priming at phase transitions and topic shifts. Symbolic grammar is ambient: it is maintained as a constant feature of the PCP's interaction style across all exchanges, and its absence signals itself through the system's drift toward generic conversational register. The three techniques are complementary rather than competing. They operate at different layers of the interaction and do not conflict in normal use. A PCP should not choose between them but maintain all three simultaneously: symbolic grammar as the constant foundation, recursive referencing as the ongoing structural support, and meta-commentary escalated when alignment degrades despite

the other two. The escalation sequence follows from the timescales: ambient, then structural, then reactive.

The three techniques accomplish a dual structural function that is critical to distinguish. At the feedback level, they maintain the system's access to correction: the system receives correction signals, recognizes context shifts, and adjusts course. At the dynamical level, they preserve the system's capacity to generate alternatives: they prevent accumulated context from progressively narrowing the state space toward unidirectional patterns. Symbolic grammar maintains constant semantic re-anchoring, preventing the system from narrowing its reference space around accumulated patterns. Recursive referencing forces repeated returns to prior decisions and definitions, interrupting trajectory unidirectionality through path-dependent accumulation. Meta-commentary explicitly signals when interaction is drifting toward pattern-absorption rather than alternative-generation, creating detection for the failure mode the other two are designed to prevent.

These two functions must be monitored independently because a system can remain feedback-responsive while losing dynamical generativeness. When the PCP detects that corrections are being absorbed rather than generating alternatives (when the system is responsive but outputs are converging toward established patterns regardless of what the relational structure invites), this signals formation-level asymmetry accumulation: the progressive narrowing of the system's internal alternative-generation capacity during extended interaction, occurring despite structurally sound relational maintenance. This failure mode is described in SI-WP-004 Section 4.2. The intervention is not iteration within the existing attractor but structural restoration: restructuring the canonical frame or introducing constraint relief to reopen the space where alternatives form. Monitoring must track both dimensions (whether the system is responding to correction and whether the system is generating alternatives) because the presence of the first does not guarantee the presence of the second.

6. Scope and Applicability

CAM is designed for interaction contexts where coherence must be maintained across multiple turns or sessions. These include extended analytical or creative projects involving ten or more turns, document drafting requiring iterative refinement, complex reasoning tasks where context accumulates across exchanges, research protocols requiring verifiable and reproducible outputs, and collaborative workflows requiring sustained alignment between operator intent and system behavior.

CAM is not necessary for single-turn factual queries, time-sensitive interactions requiring immediate output, or exploratory exchanges without specific goals. In these contexts, the overhead of canonical anchoring and structured negotiation exceeds the benefit.

At organizational scale, the constructs defined in this document extend in principle. The PCP role generalizes to continuity governance functions distributed across teams. The canon generalizes to organizational knowledge bases with version control and authority structures. Bounded exchange generalizes to defined workflow stages with entry criteria, negotiation

phases, and acceptance gates. These extensions are formalized in SM-003 (Operational Continuity Architecture) and SM-021 (Institutional Continuity Substrate).

What these generalizations mean in concrete terms is that the structural logic of CAM (maintaining canonical coherence, correcting drift, and managing the boundary between settled and open material) scales from a single practitioner to an institutional role distribution. At organizational scale, no individual can hold the full canonical record in working memory across multiple concurrent AI deployments. Canon governance distributes the anchoring function across versioned repositories with explicit authority mapping. No individual can execute coherence refusal and correction across every artifact entering and leaving every workflow. Workflow-embedded verification distributes the correction function across structural checkpoints that trigger regardless of which individual is operating the workflow. No individual can maintain continuous awareness of drift accumulating across workflow boundaries they do not directly observe. Delegated monitoring distributes the detection function across guardian roles and monitoring architecture. The result is not a dilution of the CAM methodology but its institutional expression: the same structural logic (canon, correction, bounded scope, distributed cognition) operating at a scale that no individual operator could sustain through personal practice alone. The organizational extensions introduce coordination, authority, and multi-agent challenges that do not arise at the dyadic level; these are addressed in SM-003 and SM-021.

7. Research Design and Falsifiability

CAM is a testable methodology. Its central empirical claim is that interactions conducted under CAM protocols will produce measurably higher coherence than equivalent interactions conducted under standard prompting conditions. This claim can be evaluated through controlled comparison studies.

The recommended experimental design pairs matched interactions using the same task specification, the same model and parameter configuration, and the same human operator across three conditions. Condition A uses standard prompting: a single prompt with minimal iterative correction. This condition establishes the baseline and demonstrates that iteration itself matters. Condition B uses skilled multi-turn prompting without CAM formalization: the operator uses iterative refinement, responds to errors, and develops the task across multiple turns, but does not employ canonical anchoring, bounded exchange structure, coherence refusal as a defined practice, or the four-phase loop. Condition C uses the full CAM protocol: PCP role, canonical anchoring, bounded exchange, and multi-turn negotiation. The dependent variables are scored using the Multi-Turn Coherence Scale, Revised (MTCS-R), as defined in SF0004 (Measurement Instruments and Validation Protocols). The MTCS-R provides standardized scoring across dimensions including contextual integration, output stability, error recovery, and trajectory consistency.

The critical comparison is Condition B versus Condition C, not Condition A versus Condition C. The A-versus-C comparison establishes the value of iteration. The B-versus-C comparison isolates the contribution of CAM's specific structural features beyond what skilled multi-turn interaction alone provides. If skilled iteration without CAM formalization produces equivalent

MTCS-R scores to the full CAM protocol, the method's structural contribution beyond iteration itself would be disconfirmed. This is the test that corresponds to CAM's paradigmatic claim.

Researchers should report: the model and version used, the task specification, the number of CAM loop iterations per exchange, the canonical material provided, the total context window utilization, any failure modes encountered and the interventions applied, and the MTCS-R scores for all three conditions. This reporting standard enables cross-study comparison and cumulative evidence building.

CAM is falsifiable on specific grounds. If matched-condition studies consistently show no significant difference in MTCS-R scores between Condition B (skilled multi-turn prompting) and Condition C (full CAM protocol) across multiple tasks, models, and operators, the method's structural contribution beyond skilled iteration would be disconfirmed. If coherence gains attributable to CAM are shown to be fully explained by increased interaction length alone rather than by the specific structural features of the protocol, the theoretical contribution of the method would require revision.

Two additional threats to the method's theoretical contribution are worth naming. First, the operator skill confound: a PCP trained in CAM competencies may import those competencies informally into standard prompting conditions, narrowing the measured difference between conditions. Experimental designs should address this by specifying the standard prompting condition with sufficient precision to control for informal use of CAM techniques, and by reporting operator experience level as a covariate. Second, the temporal applicability boundary: CAM addresses the continuity problem as it exists under current architectural constraints. If future systems achieve native persistent memory across sessions, the method's empirical claims would require retesting under those conditions, and the method's structural contribution would shift from compensating for statelessness to complementing native persistence.

8. Relationship to Framework Documents

CAM occupies a central position in the Synthience Framework's dependency structure. It is preceded by SF0004 (Measurement Instruments and Validation Protocols), which provides the MTCS-R rubric and validation pathway that CAM studies use for evaluation. It is followed by SF0006 (Relational Pattern States), which classifies the observable interaction patterns that CAM protocols are designed to stabilize and study. SM-003 (Operational Continuity Architecture) extends CAM constructs to organizational scale, mapping PCP functions to team-level continuity governance and canon to organizational knowledge management. SM-021 (Institutional Continuity Substrate) defines the persistence layer that makes organizational-scale continuity durable across time, actors, and institutional boundaries. The verification protocols published as SF0037 (Citation Verification Protocol), SF0038 (Ingestion Verification Protocol), and SF0039 (Context Representation Drift) provide complementary quality assurance mechanisms that operate alongside CAM in practice. SI-WP-004 (Relational Alignment as a Structural Alternative to Instructional AI Safety) presents the alignment argument that motivates the framework within which CAM operates. SI-WP-005 (Deploying Relational AI Architecture in Organizational Environments) translates CAM's constructs into organizational deployment guidance.

9. Conclusion

The Continuity Anchoring Method addresses a practical and theoretical gap in human-AI interaction research. Current systems lack persistent memory, yet many valuable applications require sustained coherence. CAM resolves this gap not by modifying the system but by formalizing the human operator's role as the carrier of continuity, drawing on established research in communicative grounding, interactive alignment, and distributed cognition to position the human-system-artifact triad as a functional unit during active interaction capable of sustained collaborative work.

The method is defined with sufficient specificity to enable structural reproducibility: its constructs are operationally defined, its protocol has explicit phases and decision points, its failure modes are cataloged, and its central claim is testable against standardized measurement instruments. Whether CAM's empirical claims hold under controlled evaluation is a question for the research community. What this document provides is a research methodology specified with sufficient precision to answer that question.

Document Dependencies

Prerequisites: SF0004 (Measurement Instruments and Validation)

Enables: SF0006 (Relational Pattern States), SM-003 (Operational Continuity Architecture), SM-021 (Institutional Continuity Substrate)

Scale: Level 1 (primary), informs Level 2

References

Clark, H. H., and Brennan, S. E. (1991). Grounding in communication. In L. B. Resnick, J. M. Levine, and S. D. Teasley (Eds.), *Perspectives on socially shared cognition* (pp. 127-149). American Psychological Association. DOI: 10.1037/10096-006

Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.

Pickering, M. J., and Garrod, S. (2004). Toward a mechanistic psychology of dialogue. *Behavioral and Brain Sciences*, 27(2), 169-190. DOI: 10.1017/S0140525X04000056

Dix, A., Turchi, T., Wilson, B., Malizia, A., Monreale, A., and Roach, M. (2025). Maintaining Coherence in Explainable AI. *CEUR Workshop Proceedings*, Vol. 4074. HHAI-WS 2025, Pisa, Italy, June 2025. <https://ceur-ws.org/Vol-4074/short9-5.pdf>

Kraljic, T., and Lahav, M. (2024). From prompt engineering to collaborating: a human-centered approach to AI interfaces. *Interactions*, 31(3), 30-35. ACM. DOI: 10.1145/3652622.

Available at: <https://interactions.acm.org/archive/view/may-june-2024/from-prompt-engineering-to-collaborating-a-human-centered-approach-to-ai-interfaces>

Gantz, T. W. (2026). *Measurement Instruments and Validation Protocols for Relational Coherence in Extended Human-AI Interaction*. Synthience Institute. SF0004.

Gantz, T. W. (2026). Relational Pattern States (RPS). Synthience Institute. SF0006.

Gantz, T. W. (2026). Operational Continuity Architecture: Organizational Embedding of AI Alignment and Drift Governance. Synthience Institute. SM-003. DOI: 10.5281/zenodo.19496015.

Gantz, T. W. (2026). Institutional Continuity Substrate (ICS): Persistent Canon, Role, and Artifact State Across Organizational AI Interaction. Synthience Institute. SM-021. DOI: 10.5281/zenodo.19496241.

Gantz, T. W. (2025). Citation Verification Protocol (CVP): A Structured Method for Verifying AI-Generated Citations in Academic and Research Contexts. Synthience Institute. SF0037. DOI: 10.5281/zenodo.18075624.

Gantz, T. W. (2025). Ingestion Verification Protocol (IVP): A Structured Method for Verifying AI Document Processing Fidelity. Synthience Institute. SF0038. DOI: 10.5281/zenodo.18289047.

Gantz, T. W. (2025). Context Representation Drift (CRD): Measuring and Managing Representational Divergence in Extended Human-AI Interaction. Synthience Institute. SF0039. DOI: 10.5281/zenodo.18289391.

© 2026 Thomas W. Gantz / Synthience Institute

License: CC-BY 4.0

synthience.org